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Tutorial Course for Intro to Sensor-Integrated Systems

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Step Response of First-order Systems

Engineering Background: Commonly used to describe components without mass or inertia, such as temperature sensors (thermocouples) in integrated sensor systems, liquid-level systems, and RC low-pass filter circuits.

- These systems only exhibit "resistance" and "capacitance" characteristics.

Core Dynamic Parameter: Time Constant τ

- **Physical Significance:** The core indicator determining the speed of the system response. The smaller the τ , the faster the sensor tracks the true signal.
- **Thermodynamic Definition:** $\tau = \frac{MC}{UA}$

Mathematical Model for Step Response:

When the system is subjected to a step input of height ΔI , the transient change in output is:

$$\Delta O(t) = K \cdot \Delta I \cdot (1 - e^{-t/\tau})$$

(Note: K is the steady-state sensitivity of the system)

Key Points:

- When $t = \tau$, the sensor output reaches **63.2%** of its final steady-state change.
- When $t = 4\tau \sim 5\tau$, the system basically reaches its new steady state.

Step Response of Second-order Systems

Engineering Background:

- Used to describe physical mechanisms containing "inertia," such as mechanical elastic force sensors and accelerometers.
- Due to the presence of mass (m), springs (k), and dampers (λ), the system is prone to overshoot and oscillation when subjected to a step impact.

Two Core Dynamic Parameters:

- **Natural Frequency ω_n :** $\omega_n = \sqrt{k/m}$. Determines the speed of the system's inherent oscillation.
- **Damping Ratio ξ :** $\xi = \frac{\lambda}{2\sqrt{km}}$. Measures the system's ability to dissipate energy and suppress oscillation (the optimal engineering design value is usually $\xi \approx 0.7$).

Mathematical Model for Underdamped ($\xi < 1$) Step Response:

$$\Delta O(t) = K \cdot \Delta I \cdot \left[1 - e^{-\xi\omega_n t} \left(\cos\omega_d t + \frac{\xi}{\sqrt{1-\xi^2}} \sin\omega_d t \right) \right]$$

(Note: Damped natural frequency $\omega_d = \omega_n \sqrt{1-\xi^2}$)

Key Points:

Waveform Characteristics: The exponential decay term $e^{-\xi\omega_n t}$ determines the **Settling Time** ($T_s \approx \frac{5}{\xi\omega_n}$), while the trigonometric terms determine the oscillation frequency.

Exercise

A temperature measurement system consists of linear elements and has an overall steady-state sensitivity of unity. The dynamics of the system are determined by the first-order transfer function of the sensing element. At time $t = 0$, the sensing element is suddenly transferred from air at $20\text{ }^{\circ}\text{C}$ to boiling water. One minute later the element is suddenly transferred back to air. Using the data given below, calculate the system dynamic error at the following times: $t = 10, 20, 50, 120$ and 300 s .

Sensor data

$$\text{Mass} = 5 \times 10^{-2} \text{ kg}$$

$$\text{Surface area} = 10^{-3} \text{ m}^2$$

$$\text{Specific heat} = 0.2 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$$

$$\text{Heat transfer coefficient for air} = 0.2 \text{ W m}^{-2} \text{ }^{\circ}\text{C}^{-1}$$

$$\text{Heat transfer coefficient for water} = 1.0 \text{ W m}^{-2} \text{ }^{\circ}\text{C}^{-1}$$

Solution

1. Calculation of Time Constants (τ)

For a first-order system, the time constant τ is calculated using:

$$\tau = \frac{mC}{hA}$$

Based on the data given in the problem:

- $m = 5 \times 10^{-2} \text{ kg}$
- $A = 10^{-3} \text{ m}^2$
- $C = 0.2 \text{ J} \cdot \text{kg}^{-1} \cdot ^\circ \text{C}^{-1}$

Time constant in boiling water (τ_w):

Given the heat transfer coefficient in water $h_w = 1.0 \text{ W} \cdot \text{m}^{-2} \cdot ^\circ \text{C}^{-1}$

$$\tau_w = \frac{(5 \times 10^{-2}) \times 0.2}{1.0 \times 10^{-3}} = \frac{0.01}{0.001} = 10 \text{ s}$$

Time constant in air (τ_a):

Given the heat transfer coefficient in air $h_a = 0.2 \text{ W} \cdot \text{m}^{-2} \cdot ^\circ \text{C}^{-1}$

$$\tau_a = \frac{(5 \times 10^{-2}) \times 0.2}{0.2 \times 10^{-3}} = \frac{0.01}{0.0002} = 50 \text{ s}$$

2. Establish the temperature response equation

The steady-state sensitivity of the measurement system is 1, which means that the measured value equals the true temperature at steady state. The dynamic error is defined as: the difference between the measured temperature and the current true environmental temperature, i.e., $E(t) = T_m(t) - T_{true}$.

Stage 1: In boiling water ($0 \leq t \leq 60 \text{ s}$)

- Initial temperature: $T_m(0) = 20^\circ \text{C}$
- Environmental temperature: $T_{\infty,1} = 100^\circ \text{C}$
- Temperature response equation:

$$T_m(t) = T_{\infty,1} + [T_m(0) - T_{\infty,1}]e^{-t/\tau_w}$$

$$T_m(t) = 100 + (20 - 100)e^{-t/10} = 100 - 80e^{-t/10}$$

- Dynamic error equation for this stage (true temperature is 100°C):

$$E_1(t) = T_m(t) - 100 = -80e^{-t/10}$$

At $t = 60 \text{ s}$ (after one minute), the temperature of the sensor is:

$$T_m(60) = 100 - 80e^{-60/10} = 100 - 80e^{-6} \approx 100 - 80 \times 0.002479 = 99.8017^\circ \text{C}$$

Solution

Stage 2: Returning to air ($t > 60$ s)

- **Initial temperature:** i.e., the temperature at the end of the first stage, $T_m(60) = 99.8017^\circ\text{C}$
- **Environmental temperature:** $T_{\infty,2} = 20^\circ\text{C}$
- **Define a new time variable:** $t' = t - 60$ (the time elapsed after returning to the air).
- **Temperature response equation:**

$$T_m(t) = T_{\infty,2} + [T_m(60) - T_{\infty,2}]e^{-(t-60)/\tau_a}$$

$$T_m(t) = 20 + (99.8017 - 20)e^{-(t-60)/50} = 20 + 79.8017e^{-(t-60)/50}$$

- **Dynamic error equation for this stage (true temperature is 20°C):**

$$E_2(t) = T_m(t) - 20 = 79.8017e^{-(t-60)/50}$$

- $t = 10$ s (Belongs to Stage 1):

$$E(10) = -80e^{-10/10} = -80e^{-1} = -80 \times 0.3679 \approx -\mathbf{29.43^\circ\text{C}}$$

- $t = 20$ s (Belongs to Stage 1):

$$E(20) = -80e^{-20/10} = -80e^{-2} = -80 \times 0.1353 \approx -\mathbf{10.83^\circ\text{C}}$$

- $t = 50$ s (Belongs to Stage 1):

$$E(50) = -80e^{-50/10} = -80e^{-5} = -80 \times 0.00674 \approx -\mathbf{0.54^\circ\text{C}}$$

- $t = 120$ s (Belongs to Stage 2):

$$E(120) = 79.8017e^{-(120-60)/50} = 79.8017e^{-60/50} = 79.8017e^{-1.2}$$

$$E(120) = 79.8017 \times 0.3012 \approx \mathbf{24.04^\circ\text{C}}$$

- $t = 300$ s (Belongs to Stage 2):

$$E(300) = 79.8017e^{-(300-60)/50} = 79.8017e^{-240/50} = 79.8017e^{-4.8}$$

$$E(300) = 79.8017 \times 0.00823 \approx \mathbf{0.66^\circ\text{C}}$$

Exercise

A force sensor has a mass of 0.5 kg, stiffness of $2 \times 10^2 \text{ N m}^{-1}$ and a damping constant of 6.0 N s m^{-1} .

- (a) Calculate the steady-state sensitivity, natural frequency and damping ratio for the sensor.
- (b) Calculate the displacement of the sensor for a steady input force of 2 N.
- (c) If the input force is suddenly increased from 2 to 3 N, derive an expression for the resulting displacement of the sensor.

Solution

(a) Calculate steady-state sensitivity, natural frequency, and damping ratio

1. Steady-state sensitivity (K):

For this type of mechanical sensor, the steady-state sensitivity is the reciprocal of the stiffness k :

$$K = \frac{1}{k} = \frac{1}{200} = \mathbf{0.005 \text{ m N}^{-1}}$$

2. Natural frequency (ω_n):

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.5}} = \sqrt{400} = \mathbf{20 \text{ rad s}^{-1}}$$

3. Damping ratio (ξ):

$$\xi = \frac{\lambda}{2\sqrt{km}} = \frac{6.0}{2\sqrt{200 \times 0.5}} = \frac{6.0}{2\sqrt{100}} = \frac{6.0}{20} = \mathbf{0.3}$$

(b) Calculate sensor displacement for a steady input force of 2 N

Under steady-state conditions, the system has no acceleration or velocity. Damping and inertial forces are zero. The initial displacement $x(0)$ can be obtained by directly multiplying the steady-state sensitivity by the input force:

$$x(0) = K \times F = 0.005 \times 2 = \mathbf{0.01 \text{ m}}$$

(c) Derive the displacement expression when the input force jumps from 2 N to 3 N

$$\Delta F = 3 - 2 = 1 \text{ N.}$$

1. Calculate the necessary intermediate parameters:

$$\Delta x_{steady} = K \times \Delta F = 0.005 \times 1 = \mathbf{0.005 \text{ m}}$$

• **Exponential decay coefficient:** $\xi\omega_n = 0.3 \times 20 = 6$

• **Damped natural frequency (actual oscillation frequency):**

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 20 \sqrt{1 - 0.3^2} = 20 \sqrt{0.91} \approx \mathbf{19.08 \text{ rad s}^{-1}}$$

• **Sine term coefficient:**

$$\frac{\xi}{\sqrt{1 - \xi^2}} = \frac{0.3}{\sqrt{0.91}} \approx \mathbf{0.314}$$

2. Write the equation for the change in displacement $\Delta x(t)$:

According to the standard formula for the step response of an underdamped system:

$$\Delta x(t) = \Delta x_{steady} \left[1 - e^{-\xi\omega_n t} \left(\cos \omega_d t + \frac{\xi}{\sqrt{1 - \xi^2}} \sin \omega_d t \right) \right]$$

Substituting the numerical values:

$$\Delta x(t) = 0.005 [1 - e^{-6t} (\cos 19.08t + 0.314 \sin 19.08t)]$$

3. Derive the final total displacement equation $x(t)$:

$$x(t) = x(0) + \Delta x(t)$$

The final expression is:

$$x(t) = 0.01 + 0.005 [1 - e^{-6t} (\cos 19.08t + 0.314 \sin 19.08t)] \quad (\text{Unit: m})$$

How to Handle Complex Periodic Signals?

Engineering Pain Point: Real-world signals are not perfect.

- In industrial settings (e.g., fluid pulsation, mechanical vibration), sensors often receive continuous periodic signals with complex waveforms rather than a single pure sine wave.
- **Fourier Series:** Any complex periodic signal can be "decomposed" into a superposition of a series of sine waves of different frequencies (Fundamental + Higher Harmonics).

General Response Model: "Deformation" of a Sine Wave

When the input is a single-frequency signal $I(t) = A_i \sin(\omega t)$, the steady-state output of a linear system must be a sine wave of the same frequency:

$$O(t) = A_i \cdot |G(j\omega)| \cdot \sin(\omega t + \phi)$$

- **Amplitude Ratio ($|G|$):** Determines whether the frequency signal is amplified or attenuated.
- **Phase Shift (ϕ):** Determines the time lag (or lead) angle of the frequency signal.

The Ultimate Problem-Solving Weapon: Superposition Principle

- Total Output = The **direct sum** of the system's responses to each individual single-frequency input (fundamental, 2nd harmonic, 3rd harmonic, etc.).

"Frequency Fingerprints" of First and Second-order Systems

First-order Systems (e.g., Thermocouples, RC Filter Circuits)

- **Amplitude Ratio:** $|G(j\omega)| = \frac{K}{\sqrt{1+(\omega\tau)^2}}$
- **Phase Shift:** $\phi = -\arctan(\omega\tau)$
- **Engineering Characteristics: Natural Low-pass Filter.** Low-frequency signals pass through with almost no loss (amplitude ratio close to 1), while high-frequency signals are severely attenuated.

Second-order Systems (e.g., Mechanical Force Transducers, Accelerometers)

- **Amplitude Ratio:** $|G(j\omega)| = \frac{K}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}}$
- **Phase Shift:** $\phi = -\arctan\left[\frac{2\xi\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right]$
- **Engineering Characteristics: Hidden Resonance Crisis.** When the input frequency ω approaches the system's undamped natural frequency ω_n and the damping ratio ξ is small, destructive **Resonance Amplification** occurs, leading to severe measurement distortion.

Exercise

A force measurement system consists of linear elements and has an overall steady-state sensitivity of unity. The dynamics of the system are determined by the second-order transfer function of the sensing element, which has a natural frequency $\omega_n = 40 \text{ rad s}^{-1}$ and a damping ratio $\xi = 0.1$. Calculate the system dynamic error corresponding to the periodic input force signal:

$$F(t) = 50(\sin 10t + \frac{1}{3} \sin 30t + \frac{1}{5} \sin 50t)$$

The complex input periodic force $F(t) = 50(\sin 10t + \frac{1}{3} \sin 30t + \frac{1}{5} \sin 50t)$ is decomposed into three independent frequency components:

- **Component 1 (Fundamental):** Frequency $\omega_1 = 10 \text{ rad s}^{-1}$, Amplitude $A_1 = 50$
- **Component 2 (3rd Harmonic):** Frequency $\omega_2 = 30 \text{ rad s}^{-1}$, Amplitude $A_2 = \frac{50}{3} \approx 16.67$
- **Component 3 (5th Harmonic):** Frequency $\omega_3 = 50 \text{ rad s}^{-1}$, Amplitude $A_3 = \frac{50}{5} = 10$

Solution

Frequency 1: $\omega_1 = 10 \text{ rad s}^{-1}$

- Frequency ratio: $\frac{\omega_1}{\omega_n} = \frac{10}{40} = 0.25$

- Amplitude ratio:

$$|G_1| = \frac{1}{\sqrt{(1 - 0.25^2)^2 + (2 \times 0.1 \times 0.25)^2}} = \frac{1}{\sqrt{0.9375^2 + 0.05^2}} \approx \mathbf{1.065}$$

- Phase shift:

$$\phi_1 = -\arctan\left(\frac{0.05}{0.9375}\right) \approx \mathbf{-3.05^\circ}$$

- Output component:

$$\underline{O_1(t) = 50 \times 1.065 \sin(10t - 3.05^\circ) = \mathbf{53.25 \sin(10t - 3.05^\circ)}}$$

Frequency 2: $\omega_2 = 30 \text{ rad s}^{-1}$

- Frequency ratio: $\frac{\omega_2}{\omega_n} = \frac{30}{40} = 0.75$

- Amplitude ratio:

$$|G_2| = \frac{1}{\sqrt{(1 - 0.75^2)^2 + (2 \times 0.1 \times 0.75)^2}} = \frac{1}{\sqrt{0.4375^2 + 0.15^2}} \approx \mathbf{2.162}$$

- Phase shift:

$$\phi_2 = -\arctan\left(\frac{0.15}{0.4375}\right) \approx \mathbf{-18.92^\circ}$$

- Output component:

$$\underline{O_2(t) = 16.67 \times 2.162 \sin(30t - 18.92^\circ) = \mathbf{36.04 \sin(30t - 18.92^\circ)}}$$

Frequency 3: $\omega_3 = 50 \text{ rad s}^{-1}$

- Frequency ratio: $\frac{\omega_3}{\omega_n} = \frac{50}{40} = 1.25$

- Amplitude ratio:

$$|G_3| = \frac{1}{\sqrt{(1 - 1.25^2)^2 + (2 \times 0.1 \times 1.25)^2}} = \frac{1}{\sqrt{(-0.5625)^2 + 0.25^2}} \approx \mathbf{1.625}$$

- Phase shift ($\triangle$ Common Error Point):

Because the real part of the denominator in the formula $1 - (\omega/\omega_n)^2 = -0.5625 < 0$, the system phase shift has already crossed the -90° boundary into the third quadrant.

You must subtract 180° from the calculation result:

$$\phi_3 = -\arctan\left(\frac{0.25}{-0.5625}\right) - 180^\circ = 23.96^\circ - 180^\circ = \mathbf{-156.04^\circ}$$

- Output component:

$$\underline{O_3(t) = 10 \times 1.625 \sin(50t - 156.04^\circ) = \mathbf{16.25 \sin(50t - 156.04^\circ)}}$$

5. Calculate System Dynamic Error

System dynamic error is defined as: the actual output signal of the measurement system minus the true input signal.

$$E(t) = [O_1(t) + O_2(t) + O_3(t)] - F(t)$$

$$\begin{aligned} E(t) = & [53.25 \sin(10t - 3.05^\circ) - 50 \sin 10t] \\ & + [36.04 \sin(30t - 18.92^\circ) - 16.67 \sin 30t] \\ & + [16.25 \sin(50t - 156.04^\circ) - 10 \sin 50t] \end{aligned}$$

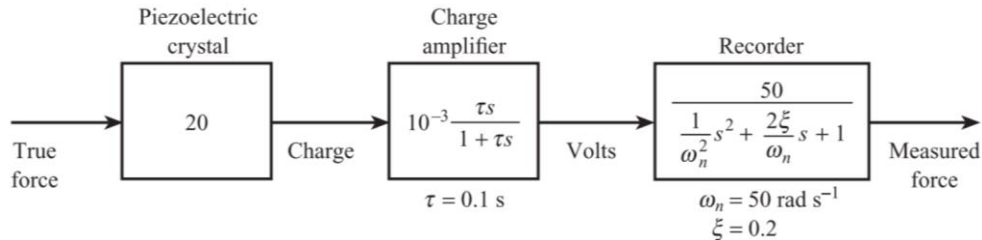
Exercise

A force measurement system consisting of a piezoelectric crystal, charge amplifier and recorder is shown in Figure Prob. 7.

- (a) Calculate the system dynamic error corresponding to the force input signal:

$$F(t) = 50(\sin 10t + \frac{1}{3} \sin 30t + \frac{1}{5} \sin 50t)$$

- (b) Explain briefly the system modifications necessary to reduce the error in (a) (hint: see Figure 8.22).



Solution

(a) Calculate the system dynamic error

1. Derive the overall transfer function $G(s)$

$$G(s) = [20] \cdot \left[10^{-3} \frac{0.1s}{1 + 0.1s} \right] \cdot \left[\frac{50}{(s/50)^2 + 2(0.2)(s/50) + 1} \right]$$

Multiplying the constants yields: $20 \times 10^{-3} \times 50 = 1$ (the static gain completely cancels out).

By substituting $s = j\omega$, we can reorganize the equation into a first-order high-pass filter (amplifier) and a second-order low-pass filter (recorder) in the frequency domain:

$$G(j\omega) = \underbrace{\left(\frac{j0.1\omega}{1 + j0.1\omega} \right)}_{\text{High-pass } H_A} \cdot \underbrace{\left(\frac{1}{\left[1 - \left(\frac{\omega}{50} \right)^2 \right] + j(0.008\omega)} \right)}_{\text{Low-pass } H_B}$$

2. Extract the components of the input signal $F(t)$

$$F(t) = 50 \sin(10t) + 16.67 \sin(30t) + 10 \sin(50t)$$

- **Fundamental component:** $\omega_1 = 10 \text{ rad s}^{-1}$, Amplitude $A_1 = 50$
- **Third harmonic:** $\omega_2 = 30 \text{ rad s}^{-1}$, Amplitude $A_2 = 16.67$
- **Fifth harmonic:** $\omega_3 = 50 \text{ rad s}^{-1}$, Amplitude $A_3 = 10$

3. Calculate the steady-state response for each frequency

Frequency 1: Fundamental $\omega_1 = 10 \text{ rad s}^{-1}$

Through High-pass H_A :

$$H_A(j10) = \frac{j1}{1 + j1} \Rightarrow |H_A| = \frac{1}{\sqrt{2}} \approx \mathbf{0.707}, \quad \angle H_A = 90^\circ - 45^\circ = \mathbf{45^\circ}$$

Through Low-pass H_B :

$$H_B(j10) = \frac{1}{[1 - 0.04] + j0.08} = \frac{1}{0.96 + j0.08} \Rightarrow |H_B| = \frac{1}{\sqrt{0.928}} \approx \mathbf{1.038},$$
$$\angle H_B \approx \mathbf{-4.76^\circ}$$

Output component $O_1(t)$:

Overall amplitude ratio = $0.707 \times 1.038 = 0.734$, Overall phase shift = $45^\circ - 4.76^\circ = 40.24^\circ$

$$O_1(t) = 50 \times 0.734 \sin(10t + 40.24^\circ) = \mathbf{36.70 \sin(10t + 40.24^\circ)}$$

Frequency 2: Third Harmonic $\omega_2 = 30 \text{ rad s}^{-1}$

Through High-pass H_A :

$$H_A(j30) = \frac{j3}{1 + j3} \Rightarrow |H_A| = \frac{3}{\sqrt{10}} \approx \mathbf{0.949}, \quad \angle H_A = 90^\circ - \arctan(3) \approx \mathbf{18.43^\circ}$$

Through Low-pass H_B :

$$H_B(j30) = \frac{1}{[1 - 0.36] + j0.24} = \frac{1}{0.64 + j0.24} \Rightarrow |H_B| = \frac{1}{\sqrt{0.4672}} \approx \mathbf{1.463}$$
$$\angle H_B = -\arctan\left(\frac{0.24}{0.64}\right) \approx \mathbf{-20.56^\circ}$$

Output component $O_2(t)$:

$$O_2(t) = 16.67 \times 1.388 \sin(30t - 2.13^\circ) = \mathbf{23.14 \sin(30t - 2.13^\circ)}$$

Solution

Frequency 3: Fifth Harmonic $\omega_3 = 50 \text{ rad s}^{-1}$ (Resonance Point)

Through High-pass H_A :

$$H_A(j50) = \frac{j5}{1 + j5} \implies |H_A| = \frac{5}{\sqrt{26}} \approx \mathbf{0.981}$$

$$\angle H_A = 90^\circ - \arctan(5) \approx \mathbf{11.31^\circ}$$

Through Low-pass H_B (Real part becomes zero):

$$H_B(j50) = \frac{1}{[1 - 1] + j0.4} = \frac{1}{j0.4} \implies |H_B| = \mathbf{2.5}, \quad \angle H_B = \mathbf{-90^\circ}$$

Output component $O_3(t)$:

$$O_3(t) = 10 \times 2.453 \sin(50t - 78.69^\circ) = \mathbf{24.53 \sin(50t - 78.69^\circ)}$$

4. Assemble the dynamic error equation

The dynamic error is defined as $E(t) = [O_1(t) + O_2(t) + O_3(t)] - F(t)$:

$$E(t) = [36.70 \sin(10t + 40.24^\circ) - 50 \sin 10t]$$

$$+ [23.14 \sin(30t - 2.13^\circ) - 16.67 \sin 30t]$$

$$+ [24.53 \sin(50t - 78.69^\circ) - 10 \sin 50t]$$

Question: Briefly explain the system modifications necessary to reduce the error in (a).

Based on the calculated steady-state frequency response of the system to the input signal, the measurement system exhibits severe low-frequency attenuation (the fundamental amplitude ratio is only 0.734) and intense high-frequency resonance distortion (the fifth harmonic amplitude ratio reaches 2.453) within its operating frequency band ($10 \text{ rad/s} \sim 50 \text{ rad/s}$).

To minimize the system's dynamic error and achieve distortion-free measurement, the following two core hardware modifications are strictly necessary:

1. Modification for Low-Frequency Attenuation: Increase the Charge Amplifier's Time

Constant τ

- **Principle Analysis:** The significant signal attenuation and phase lead occurring at $\omega_1 = 10 \text{ rad/s}$ are due to the charge amplifier acting as a first-order high-pass filter. Its current time constant $\tau = 0.1 \text{ s}$ results in a high-pass cutoff frequency ($\omega_c = 1/\tau = 10 \text{ rad/s}$) that is too high, which severely interferes with the effective operating bandwidth.
- **Modification Scheme:** The time constant τ of the charge amplifier must be significantly increased (which can be achieved physically by increasing the feedback capacitance or resistance in the circuit). This will shift its high-pass cutoff frequency to a value well below the lowest frequency component of the input signal (the fundamental 10 rad/s), thereby ensuring that low-frequency signals remain within the flat passband where the amplitude ratio is constantly 1.

Solution

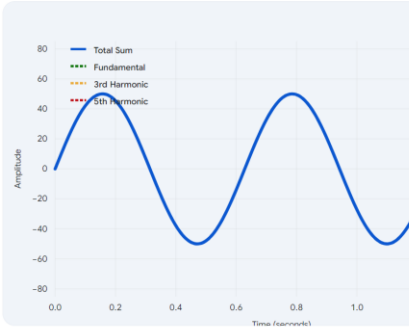
2. Modification for High-Frequency Resonance: Optimize the Recorder's Damping Ratio ξ and Natural Frequency ω_n

- **Principle Analysis:** The severe amplitude amplification and phase lag occurring at $\omega_3 = 50 \text{ rad/s}$ are due to this high-frequency component directly exciting the undamped natural frequency ($\omega_n = 50 \text{ rad/s}$) of the recorder (a second-order low-pass system). Additionally, the current damping ratio ($\xi = 0.2$) is too low, leaving the system in a severely underdamped state that cannot effectively suppress the resonance peak.
- **Modification Scheme:**
 - **(Primary Requirement):** Mechanical or electromagnetic damping must be added to the recorder to increase its damping ratio ξ to the optimal dynamic damping value (approximately 0.7). This will completely eliminate the amplitude distortion caused by the resonance peak.
 - **(Optimization Requirement):** Provided the physical structure allows, the undamped natural frequency ω_n of the recorder should be further increased (e.g., by increasing mechanical stiffness or reducing the mass of the moving parts) so that it is significantly higher than the highest frequency component of the input signal (the fifth harmonic 50 rad/s). This will effectively broaden the usable flat bandwidth of the measurement system.

Solution

Fourier Series Synthesis

PEAK AMPLITUDE	COMPLEXITY
50.0	1 components



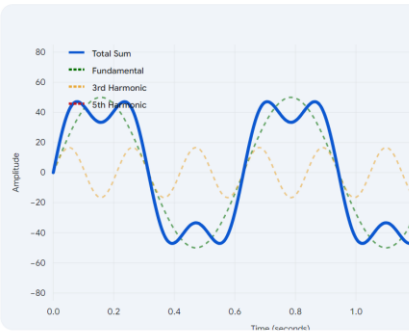
Fundamental (10 rad/s) ☒

3rd Harmonic (30 rad/s) ☐

5th Harmonic (50 rad/s) ☐

Fourier Series Synthesis

PEAK AMPLITUDE	COMPLEXITY
47.1	2 components



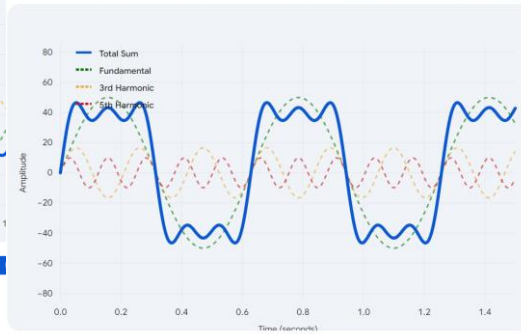
Fundamental (10 rad/s) ☒

3rd Harmonic (30 rad/s) ☒

5th Harmonic (50 rad/s) ☐

Fourier Series Synthesis

PEAK AMPLITUDE	COMPLEXITY
46.7	3 components



Fundamental (10 rad/s) ☒

3rd Harmonic (30 rad/s) ☒

5th Harmonic (50 rad/s) ☒

Open-loop Dynamic Compensation

Core Philosophy: Mathematical Cancellation

- When the physical characteristics of a sensor are poor (e.g., the time constant τ is too large, leading to a sluggish response) and cannot be changed physically, we can series-connect an **electronic compensation circuit** (such as a lead-lag network) at its output.
- **Zero-Pole Cancellation:** Design the transfer function of the circuit to generate a "zero" that precisely cancels out the "pole" of the original sensor, while introducing a new pole with a much smaller time constant.

Mathematical Expression:

- Original Sensor: $G_s(s) = \frac{1}{1+\tau_{slow}s}$
- Compensation Circuit: $G_c(s) = \frac{1+\tau_{slow}s}{1+\tau_{fast}s}$
- Total Transfer Function: $G_{total}(s) = G_s(s) \cdot G_c(s) = \frac{1}{1+\tau_{fast}s}$

Engineering Pros & Cons:

- **Pros:** Low cost and simple to implement; significantly broadens the flat bandwidth of the system.
- **Fatal Weakness: Extremely poor anti-interference capability.** If environmental changes cause the sensor's physical τ_{slow} to drift (parameter mismatch), the "perfect cancellation" will fail completely.

Closed-loop Negative Feedback Architecture

Core Philosophy: The Ultimate Architectural Revolution

- Stop letting the sensor "fight alone." Instead, introduce a feedback loop that converts the output signal back into a physical quantity at the input, comparing it in real-time with the original input (finding the error). The system then only responds to the "net error."

A Leap in Engineering Performance (Example: Force-Balanced Accelerometer):

- **Explosive Bandwidth & Speed:** The natural frequency of the closed-loop system ω_{nc} is much higher than that of the open-loop sensor ω_n .
- **Perfect Dynamic Damping:** By adjusting the circuit magnification, the closed-loop damping ratio ξ_c can be locked at the optimal **0.7**, completely eliminating annoying mechanical oscillations.
- **Ultimate Robustness:** Steady-state sensitivity K is no longer affected by aging mechanical parameters (like spring stiffness); instead, it depends on highly stable electronic components (standard resistors) and magnetic field strength.

Exercise

An uncompensated thermocouple has a time constant of 10 s in a fast-moving liquid.

- (a) Calculate the bandwidth of the thermocouple frequency response.
- (b) Find the range of frequencies for which the amplitude ratio of the uncompensated thermocouple is flat within $\pm 5\%$.
- (c) A lead/lag circuit with transfer function $G(s) = (1 + 10s)/(1 + s)$ is used to compensate for thermocouple dynamics. Calculate the range of frequencies for which the amplitude ratio of the compensated system is flat within $\pm 5\%$.
- (d) The velocity of the liquid is reduced, causing the thermocouple time constant to increase to 20 s. By sketching $|G(j\omega)|$ explain why the effectiveness of the above compensation is reduced.

Solution

The uncompensated thermocouple (a first-order system) has a time constant $\tau = 10$ s.

Assuming its static sensitivity $K = 1$, the original transfer function is:

$$G_u(s) = \frac{1}{1 + 10s}$$

(a) Calculate the bandwidth of the thermocouple frequency response.

For a first-order system, the bandwidth is defined as the frequency at which the amplitude ratio drops to $1/\sqrt{2} \approx 0.707$ (i.e., the -3dB cutoff frequency ω_c).

The formula is:

$$\omega_c = \frac{1}{\tau}$$

Substituting the given data:

$$\omega_c = \frac{1}{10} = \mathbf{0.1 \text{ rad/s}}$$

(b) Find the range of frequencies for which the amplitude ratio of the uncompensated thermocouple is flat within $\pm 5\%$.

A first-order low-pass system only attenuates high-frequency signals, so its maximum amplitude ratio is 1. An error within $\pm 5\%$ means the amplitude ratio $|G_u(j\omega)|$ must remain greater than or equal to **0.95**.

Setting up the inequality for the uncompensated system:

$$|G_u(j\omega)| = \frac{1}{\sqrt{1 + (10\omega)^2}} \geq 0.95$$

Solving for ω :

$$1 + 100\omega^2 \leq \left(\frac{1}{0.95}\right)^2 \approx 1.108$$

$$100\omega^2 \leq 0.108 \implies \omega^2 \leq 0.00108$$

$$\omega \leq \mathbf{0.0329 \text{ rad/s}}$$

Solution

(c) Calculate the range of frequencies for which the amplitude ratio of the compensated system is flat within $\pm 5\%$.

When the lead/lag circuit with the transfer function $G_c(s) = (1 + 10s)/(1 + s)$ is added, the overall transfer function becomes the product of the two.

The term $(1 + 10s)$ in the compensator's numerator perfectly cancels out the slow pole in the sensor's denominator (a technique known as **Zero-Pole Cancellation**):

$$G_{comp}(s) = G_u(s) \cdot G_c(s) = \left(\frac{1}{1 + 10s} \right) \cdot \left(\frac{1 + 10s}{1 + s} \right) = \frac{1}{1 + s}$$

The entire system now behaves as a new first-order system with an improved time constant

$\tau_{new} = 1$ s. Recalculating the 0.95 amplitude threshold:

$$|G_{comp}(j\omega)| = \frac{1}{\sqrt{1 + (1\omega)^2}} \geq 0.95$$

$$1 + \omega^2 \leq 1.108 \implies \omega^2 \leq 0.108$$

$$\omega \leq \mathbf{0.329} \text{ rad/s}$$

Solution

1. Root Cause: "Pole-Zero Mismatch" Induced by Parameter Drift

- **The Gap Between Theory and Reality:** The original compensation design (the green curve) relies heavily on a perfect mathematical cancellation (the circuit's zero precisely negating the sensor's pole, assuming both have $\tau = 10$ s).
- **Physical Condition Shift:** When the fluid velocity decreases, the sensor's actual thermal inertia increases (τ becomes 20 s). The fixed, hard-coded circuit parameters cannot adapt to this change, causing the mathematical cancellation to fail completely.

2. Visual Characteristics: "Mid-Frequency Dip" and Bandwidth Reduction

- **The Dip:** As illustrated by the red curve, the system encounters the actual, slower pole ($1/20$ rad/s) prematurely, which initiates attenuation. Meanwhile, the compensator's zero ($1/10$ rad/s) lags in providing the necessary amplification. This timing discrepancy creates a massive "dip" (sag) in the mid-frequency range ($0.05 \sim 0.3$ rad/s).
- **Breaking the Threshold:** The red curve rapidly drops below the 0.95 dashed line (5% error limit) at a very low frequency. This indicates that the system's high-accuracy flat bandwidth is severely compressed, resulting in performance that is arguably worse than the original, uncompensated baseline in that mid-frequency band.

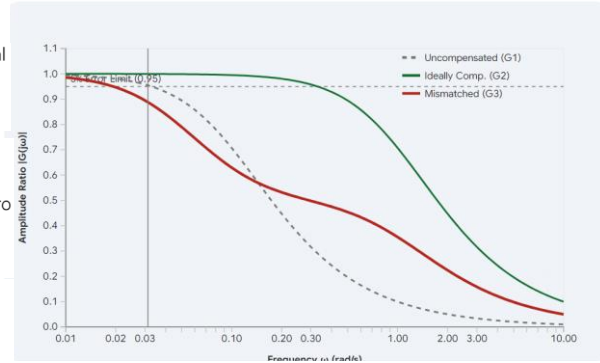
3. Engineering Insight: The "Fragility" of Open-Loop Compensation

- While open-loop dynamic compensation (lead-lag network) is low-cost and yields impressive theoretical results, it fundamentally lacks **robustness**.
- It demonstrates that relying solely on static circuit compensation is highly unreliable in real-world industrial environments where external disturbances are common and plant parameters are prone to fluctuation.

Thermocouple Frequency Response

FREQUENCY (W)	G1 RATIO	G2 RATIO	G3 RATIO
0.03 rad/s	0.954	1.000	0.888

Hover over the chart to inspect Amplitude Ratios at specific frequencies.



Exercise

An elastic force sensor has an effective seismic mass of 0.1 kg, a spring stiffness of 10 N m^{-1} and a damping constant of 14 N s m^{-1} .

- (a) Calculate the following quantities:
 - (i) sensor natural frequency
 - (ii) sensor damping ratio
 - (iii) transfer function relating displacement and force.
- (b) The above sensor is incorporated into a closed-loop, force balance accelerometer. The following components are also present:

Potentiometer displacement sensor: sensitivity 1.0 V m^{-1}

Amplifier: voltage input, current output, sensitivity 40 A V^{-1}

Coil and magnet: current input, force output, sensitivity 25 N A^{-1}

Resistor: $250 \text{ }\Omega$.

- (i) Draw a block diagram of the accelerometer.
- (ii) Calculate the overall accelerometer transfer function.
- (iii) Explain why the dynamic performance of the accelerometer is superior to that of the elastic sensor.